5.3.5 Embedded voltage reference

The parameters provided in Table 22. Embedded internal voltage reference are derived from tests performed under the ambient temperature and supply voltage conditions summarized in Section 5.3.1.

Table 22. Embedded internal voltage reference

The values in this table are specified by design and not tested in production.

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
V_{REFINT}	Internal reference voltages	-40°C < T_J < 140 °C	1.180	1.217	1.250	V
$t_{S\_vrefint}$ (1)(2)	ADC sampling time when reading the internal reference voltage	-	4.3	-	-	µs
t_{S_vbat} (1)	VBAT sampling time when reading the internal VBAT reference voltage	-	9	-	-	µs
$t_{start\_vrefint}$ (2)	Start time of reference voltage buffer when ADC is enable	-	-	-	4.4	µs
I_{refbuf} (2)	Reference Buffer consumption for ADC	V_{DDA} = 3.3 V	9	13.5	23	µA
ΔV_{REFINT} (2)	Internal reference voltage spread over the temperature range	-40 °C < T_J < 140 °C	-	5	15	mV
T_{coeff}	Average temperature coefficient	Average temperature coefficient	-	19	67	ppm/°C
$V_{DDcoeff}$	Average Voltage coefficient	3.0 V < V_{DD} < 3.6 V	-	10	1370	ppm/V

1. *The shortest sampling time for the application can be determined by multiple iterations.*
2. *Specified by design - Not tested in production.*

Table 23. Internal reference voltage calibration values

Symbol	Parameter	Memory address
V_{REFINT_CAL}	Raw data acquired at temperature of 30 °C, V_{DDA} = 3.3 V	0x08FFF810 - 0x08FFF812

5.3.6 Supply current characteristics

The current consumption is a function of several parameters and factors such as the operating voltage, ambient temperature, I/O pin loading, device software configuration, operating frequencies, I/O pin switching rate, program location in memory and executed binary code.

The current consumption is measured as described in Current consumption measurement.

Typical and maximum current consumption

The MCU is placed under the following conditions:

- All I/O pins are in analog input mode.
- All peripherals are disabled except when explicitly mentioned.
- The flash memory access time is adjusted with the minimum wait-state number, depending on the f_{HCLK} frequency (refer to the tables *Number of wait states according to CPU clock (HCLK) frequency* available in the product reference manual).
- When the peripherals are enabled, $f_{PCLK} = f_{HCLK}$.

The parameters given in the tables below are derived from tests performed under ambient temperature and supply voltage conditions summarized in Section 5.3.1: General operating conditions. If not specified otherwise, typical data are measured with a V_{DD} supply of 3.0 V, and maximum data are measured at 3.6 V.